Introduction to Large Language Models
Assignment- 9

TYPE OF QUESTION: MCQ
Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

Which statement best explains why a knowledge graph (KG) is a more flexible representation than a traditional relational knowledge base?

a. A KG needs no schema at all, whereas a relational knowledge base always does.
b. A KG stores entities as nodes joined by typed edges, so heterogeneous relations and multi-hop traversal are represented directly.
c. A KG stores every fact as a hypergraph, which removes all redundancy.
d. A KG makes textual sources unnecessary because it stores every possible fact.

Correct Answer: b

Explanation:

A relational knowledge base fixes the facts into tables with predefined columns, so a new kind of relation usually means a new table or schema change. A KG represents each fact as a (subject, relation, object) triple, which lets new relation types be added as new edge labels and lets queries follow chains of edges for inference.
(a) overstates the point: knowledge graphs commonly do have an ontology or schema, it is simply more permissive. (c) and (d) are false - a KG is neither restricted to hypergraphs nor complete.

QUESTION 2: [1 mark]

Entity and relation alignment across knowledge graphs in different languages is a hard problem. Which of the following genuinely contribute to effective alignment?

a. Transliteration and cross-lingual string matching over entity labels
b. Comparing the neighbourhood structure of candidate entities in the two graphs
c. Requiring that every alias of an entity appears identically in both graphs
d. Matching relations purely on the lexical similarity of their names, ignoring context

Correct Answers: a, b

Explanation:

(a) True: names often survive translation in transliterated form, so string-level matching over labels is a useful first signal.
(b) True: structure disambiguates what labels cannot. If two candidates share neighbours that are themselves already aligned, that is strong evidence they denote the same entity.
(c) False: the whole point of alignment is to link differing surface forms, not to force them to be identical.
(d) False: lexically similar relation names can mean different things across languages, so context and the types of entities they connect must be considered.

QUESTION 3: [1 mark]

Which of the following are learned jointly in an end-to-end differentiable KGQA system, as opposed to a semantic parsing approach?

a. A dense representation of the natural language question
b. Embeddings of the entities and relations in the graph

c. A hand-specified logical form for each question template

d. The scoring mechanism that selects which node answers the question

Correct Answers: a, b, d

Explanation:

A differentiable KGQA model encodes the question, encodes the graph, and learns to score candidate answer nodes, with gradients flowing through all three. Nothing in the pipeline is symbolic.

(c) belongs to semantic parsing, which converts the question into an explicit logical form and executes it against the graph. That route is more interpretable but needs grammars or annotated logical forms; the differentiable route is trainable end to end but harder to inspect.

QUESTION 4: [1 mark]

Which statements correctly describe the local closed-world assumption used when generating negatives for KG training?

a. Triples not present in the graph are treated as false during training.

b. It is strictly correct, because knowledge graphs are complete by construction.

c. Some sampled negatives are in fact missing true triples, so the training signal carries label noise.

d. It removes the need for a separate validation split.

Correct Answers: a, c

Explanation:

Knowledge graphs record only positive facts, so training needs an assumption to manufacture negatives. The local closed-world assumption supplies one: anything not stated is taken as false.

(b) is precisely what the assumption is not - KGs are known to be incomplete, which is the reason KG completion exists as a task. (c) follows directly: a corrupted triple may happen to be a true fact that is simply absent from the graph. (d) is unrelated to how negatives are formed.

QUESTION 5: [1 mark]

A KG completion model is trained with a softmax over entities to model P(o | s, r). Which of the following statements about replacing the full softmax with a sampled one are correct?

a. The exact denominator requires scoring every entity in the graph, which is prohibitively expensive for large KGs.

b. The true object must be kept in the sampled denominator, otherwise the estimate is biased.

c. Sampling negatives uniformly tends to produce easy negatives, giving a high-variance training signal.

d. Sampled softmax removes the need to learn entity embeddings.

Correct Answers: a, b, c

Explanation:

(a) True: the normaliser runs over the whole entity set, so its cost grows with the size of the graph and dominates training.

(b) True: the sampled denominator is meant to approximate the full one, and dropping the true object from it systematically distorts the ratio.

(c) True: a randomly drawn entity is usually obviously wrong, so most sampled negatives carry little gradient signal. Many samples are then needed before a useful, borderline negative appears, which is why hard or self-adversarial negative sampling was introduced.

(d) False: the embeddings are exactly what is being learned; only the normalisation is approximated.

QUESTION 6: [1 mark]

A model is trained with the pairwise hinge loss max{ 0, m + f(s', r, o') - f(s, r, o) }, where (s, r, o) is a positive triple, (s', r, o') a sampled negative, m > 0 the margin, and a higher f means a more plausible triple. The loss is strictly positive when:

a. f(s, r, o) >= f(s', r, o') + m
b. f(s, r, o) < f(s', r, o') + m
c. f(s', r, o') >= f(s, r, o) + m
d. only when m = 0

Correct Answer: b

Explanation:

The hinge is active whenever m + f(negative) - f(positive) > 0, that is whenever the positive triple fails to outscore the negative by at least the margin m.
(a) is the complementary condition and gives exactly zero loss - the constraint is already satisfied, so that pair contributes no gradient. (c) is a strictly stronger violation, and it is covered by (b) rather than being the boundary. (d) is false, since the loss behaves this way for any positive margin.

QUESTION 7: [1 mark]

For a single test query, a KG completion model ranks the correct entity 4th out of 100 candidates. What are hits@1, hits@5 and the reciprocal rank for this query?

a. 0, 1, 0.25
b. 1, 1, 4
c. 0, 0, 0.25
d. 0, 1, 4

Correct Answer: a

Explanation:

hits@K is 1 when the correct entity falls within the top K of the ranked list and 0 otherwise. Rank 4 is outside the top 1, so hits@1 = 0, and inside the top 5, so hits@5 = 1.
The reciprocal rank is 1 / rank = 1 / 4 = 0.25. Averaging that quantity over all test queries gives the MRR, which is why MRR rewards moving the correct answer from rank 4 to rank 2 far more than from rank 40 to rank 20.

QUESTION 8: [1 mark]

Which statements about filtered evaluation for KG completion are correct?

a. Candidate entities that form known true triples in the training or validation data are removed from the ranked list before the test entity is scored.
b. It avoids penalising the model for placing a different, genuinely correct answer above the held-out one.
c. It can only ever lower the reported score compared with the raw setting.
d. It changes ranking metrics such as MRR and hits@K.

Correct Answers: a, b, d

Explanation:

Many KG relations are one-to-many. For the query (Delhi, located_in, ?) several objects are correct, and some of them appear in the training set. Unfiltered ranking counts those as errors ahead of the test answer, which understates model quality.
(c) is the reverse of what happens: removing known-correct distractors can only move the test entity up the list, so filtered scores are at least as high as raw scores.

QUESTION 9: [1 mark]

Why can TransE not represent a non-trivial symmetric relation?

a. Because the norm of every relation embedding in TransE is constrained to 1.

b. Because symmetry would require both $s + r \approx o$ and $o + r \approx s$, which forces $r \approx 0$ and collapses the relation.

c. Because the subject and object embeddings would have to be orthogonal.

d. Because TransE uses complex-valued embeddings that cannot encode symmetry.

Correct Answer: b

Explanation:

TransE scores a triple by the distance $\| s + r - o \|$, so a valid triple satisfies $s + r \approx o$. If the relation is symmetric, the reverse triple must also be valid, giving $o + r \approx s$. Adding the two conditions gives $2r \approx 0$, so the only symmetric relation TransE can represent is the zero vector, which then makes every pair of identical entities equally plausible.

(d) is false - TransE embeddings are real-valued; it is RotatE that moves to the complex plane, which is precisely how it recovers symmetry.

QUESTION 10: [1 mark]

Which of the following is the RotatE scoring function?

a. $f(s, r, o) = \| s + r - o \|$

b. $f(s, r, o) = \| s \odot r - o \|$, where every component of r has modulus 1

c. $f(s, r, o) = s^T M_r o$, with M_r a full matrix

d. $f(s, r, o) = - < s, r, o >$, the real part of a three-way inner product

Correct Answer: b

Explanation:

RotatE embeds entities and relations in the complex plane and treats a relation as an element-wise rotation. Constraining each component of r to modulus 1 means the operation changes only the phase, so the relation is a pure rotation and the score is the distance between the rotated subject and the object.

This is what buys the extra expressiveness: a relation is symmetric exactly when $r \odot r = 1$, that is when every rotation angle is 0 or pi; an inverse relation is the rotation in the opposite direction; and composition is the sum of the angles.

(a) is TransE, (c) is the general bilinear form used by RESCAL, and (d) is the ComplEx-style trilinear score.
